

DINESH

AI/ML Engineer

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Professional Summary

AI/ML Engineer with hands-on experience building **production-grade LLM applications, RAG pipelines, and Text-to-SQL systems**. Skilled in designing scalable **FastAPI-based backend systems**, integrating **OpenAI APIs, vector databases (Pinecone, Qdrant)**, and implementing **LLMOps practices** including caching, observability, and CI/CD. Proven ability to reduce latency and cost through optimized architectures, with experience deploying **serverless AI systems on AWS**. Strong foundation in **machine learning, NLP, and system design**.

Technical Skills

LLM & Generative AI: OpenAI API (GPT-4/5), LangChain, LangGraph, RAG Pipelines, Prompt Engineering, Fine-tuning (Flan-T5, Llama, Mistral), Hugging Face Transformers, Embeddings, NLP, Chatbot Development

Vector Databases: Pinecone, Qdrant, FAISS, Chroma — semantic search, document ingestion, chunking strategies

Backend & APIs: FastAPI, Flask, RESTful APIs, Async Programming, Swagger/OpenAPI

Databases & Caching: PostgreSQL, MySQL, Redis (Upstash)

MLOps & LLMOps: Docker, CI/CD (GitHub Actions), AWS Lambda, ECR, Experiment Tracking (W&B), LLM Observability

Cloud & Tools: AWS (EC2, ECR, Lambda, App Runner), Azure (familiar), Git, GitHub, Linux

Programming: Python

Projects

Multi-Source RAG + Text-to-SQL System LLMOps	GitHub
- Built a production-grade system routing queries between RAG and Text-to-SQL pipelines for unified access to structured + unstructured data. - Developed FastAPI backend (18+ async endpoints) with caching (Redis) achieving 40–60% cost reduction and 5–10ms latency for cached responses. - Integrated OpenAI GPT-4, Pinecone for semantic retrieval and Vanna.ai for deterministic SQL generation. - Deployed via AWS Lambda + Docker + CI/CD (GitHub Actions) with real-time LLM observability.	
Flan-T5 Text Summarization NLP Fine-tuning & MLOps	GitHub
- Fine-tuned google/flan-t5-base (open-source LLM) on SAMSum dataset using Hugging Face Seq2SeqTrainer — adapting the model for dialogue summarization tasks. - Built FastAPI inference service for real-time summarization with optimized async model loading pipeline. - Integrated Weights & Biases (W&B Weave) for experiment tracking, inference logging, and model evaluation. - Containerized (Docker, linux/amd64) and deployed on AWS App Runner via ECR with GitHub Actions CI/CD for automated testing, building, and deployment.	
Drinks Quality Prediction System ML & MLOps Project	GitHub
- Designed end-to-end ML system following Experiment → Production workflow: data ingestion, validation, transformation, training, and evaluation stages. - Built RESTful prediction APIs using Flask; containerized with Docker and deployed on AWS (EC2, ECR) with CI/CD via GitHub Actions.	

Internship Experience

Data Analyst Intern AXL - Zenix	<i>Jun 2025 – Jul 2025</i>
- Analyzed structured datasets using Python (Pandas, NumPy) and SQL to surface actionable business insights and support data-driven decisions. - Performed data cleaning, preprocessing, and exploratory data analysis (EDA) to improve data quality for downstream ML workflows. - Wrote optimized SQL queries for data extraction, transformation, and reporting across multiple business domains.	

Education

Kurukshetra University	<i>Aug 2024 – May 2026</i>
<i>M.Sc. in Statistics – Coursework: Time-Series Forecasting, Probability, Statistical Inference</i>	
Deenbandhu Chhotu Ram University of Science & Technology, Murthal	<i>Aug 2020 – Jun 2023</i>
<i>B.Sc. in Mathematics – Coursework: Probability, DBMS, Linear Algebra, Calculus</i>	